

COURTESY OF SILAS OTIENO

ECONOMICS OF INFORMATION TECHNOLOGY

1. INTRODUCTION

1.1 The role of ICT in the economy

The recent global financial crisis has laid to rest several myths regarding the new economy: the business cycle is not dead, stock market valuations must be realistic and backed by sound profit expectations, and the ICT sector is not immune to downturns.

But this should not distract from the economic benefits that have already accompanied the spread of ICT and the continued importance of ICT for growth in the years to come. It may be too early to tell how the role of ICT in growth and productivity performance will develop in the first decade of the 21st century.

Some general trends can be observed, however, that suggest that ICT will continue to be a driver of growth:

- i) Productivity growth in the United States is led by ICT. Growth and productivity improvements, has continued to be strong during the recent slowdown, suggesting that part of the acceleration in productivity growth over the second half of the 1990s was indeed structural. Productivity growth in Australia and Canada, both countries characterized by ICTintensive growth, has also been strong over the recent past.
- ii) ICT networks have now spread throughout much of the OECD business sector, and will increasingly be made to work to enhance productivity and business performance. Technological progress in ICT goods and services is continuing at a rapid pace, driving prices down and leading to a wide range of new applications. For example, business-to-consumer ecommerce continues to gain in importance,

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broadband is diffusing rapidly, and activity in the telecommunications sector continues to grow. Moreover, several applications, such as broadband and e-commerce, are still in their early stages and may have a large potential for future growth.

- iii) While ICT investment has dropped off during the recent slowdown, the release of increasingly powerful microprocessors is projected to continue for the foreseeable future, which will encourage ICT investment and support further productivity growth. Nevertheless, the level of ICT investment may well be lower than that observed prior to the slowdown, however, as the 1995-2000 period was characterized by some one-off investment peaks, e.g. investments related to Y2K and the diffusion of the Internet. On the other hand, some countries may still have scope for catch-up; by 2000, Japan and the European Union area invested a similar share of total investment in ICT than the United States did in 1980.
- iv) Further technological progress in ICT production will imply a continued positive contribution of the ICT manufacturing sector to multifactor productivity (MFP) growth, notably in countries with large ICT-producing sectors such as Finland, Ireland, Japan, Korea, Sweden and the United States.

Three impacts of ICT on economic growth can be distinguished.

- i) Investment in ICT adds to the capital stock that is available for workers and thus helps raise labour productivity. ICT investment accounted for between 0.3 and 0.8 percentage point of growth in world's GDP per capita over the 1995-2001 period. The United States, Canada, the Netherlands and Australia received the largest boost; Japan and the United Kingdom a more modest one, and Germany, France and Italy a much smaller one. Investment in software accounted for up to a third of the contribution of ICT capital.
- ii) The ICT-producing sector plays an important role in some OECD countries, although it is small in most. Having an ICT-producing sector

can be important, since it has been characterized by rapid technological progress and very strong demand. In Finland, Ireland and Korea, close to 1 percentage point of aggregate economic growth in the 1995-2001 period was due to ICT manufacturing. In the United States, Japan and Sweden, the ICT producing sector also contributed significantly to productivity growth.

- iii) New evidence from an OECD-led consortium of researchers in 13 OECD countries demonstrates that the use of ICT throughout the value chain contributes to improved firm performance. The smart use of ICT can help firms increase their overall efficiency in combining labour and capital, or multi-factor productivity (MFP).
- iv) ICT use can also contribute to network effects, such as lower transaction costs and more rapid innovation, which can improve MFP. In some countries, notably the United States and Australia, there is evidence that sectors that have invested most in ICT, such as wholesale and retail trade, have experienced more rapid MFP growth. Firm-level studies also show that the use of ICT may help efficient firms gain market share at the cost of less productive firms, raising overall productivity. In addition, the use of ICT may help firms expand their product range, customize the services offered, or respond better to demand, i.e. to innovate. Moreover, ICT may help reduce inventories or help firms integrate activities throughout the value chain. These studies also show that ICT is part of a broader range of changes that help enhance performance. The impacts of ICT depend on complementary investments, e.g. in appropriate skills, and on organizational changes, such as new strategies, new business processes and new organizational structures. Firms adopting these practices tend to gain market share and enjoy higher productivity gains than other firms. ICT use by firms is also closely linked to the ability of a company to adjust to changing demand and to innovate. Users of ICT often help make their investments more valuable through their own experimentation and innovation, e.g. the introduction of new processes, products and applications.

1.2 ICT and Vision 2030

The Vision 2030 has identified Business Process Outsourcing (BPO) as an emerging and growing sector expected to become the sector of choice for employment among the youth and young professionals. The country will thus quickly establish the necessary capacity for BPO to flourish and catch up with other destinations in Africa.

Business Process Outsourcing (BPO) involves the transfer of non-core business processes along with the associated operational activities and responsibilities to a third party with at least a guaranteed equal service level.

a) Situation Analysis, Emerging Issues and Challenges

- ✦ In Kenya, the BPO sector is a small, new, but growing part of the economy. It currently account for less than 0.01 percent of GDP, 500 seats and 1,000 employees. In a day, one seat can support more than one worker depending on the number of shifts per day. The global BPO sector however is large and growing rapidly. Kenya plans to take a portion of this market. In doing so it has a comparative advantage due to quality education, volume of people trained in various foreign language skills and a time zone that aligns her to those of major global economies. In this regard, the sector will be targeted to ensure Kenya diversifies her service and product offering in the global market.
- ✦ The government has in the past undertaken various measures to improve the BPO sector performance. Key among them is development of a comprehensive national ICT policy launched in 2006 with a main objective of making Kenya an ICT hub and a premier location for Business Process Outsourcing (BPO) in Africa, the establishment of the ICT board and the launching of the East African Marine Cable Systems (TEAMS) project among others.

Key challenges facing the sector are:

- i) Poor telecommunication infrastructure leading to high costs. Data transmission costs in Kenya are three times more expensive than those of its competitors;

- ii) The high cost and unreliability of energy supply. For instance, energy costs are twice as expensive as those of other BPO destinations, such as India and the Philippines.
- iii) Lack of dedicated BPO facilities which weaken the attractiveness of Kenya as a primary BPO destination.
- iv) Inadequate supply of necessary skills.
- v) Poor local supplier base of the necessary hardware and software
- vi) No targeted incentives for BPO apart from those at the standard Export Processing Zones (EPZ)

b) BPO Incentive Framework

A comprehensive set of incentives will be designed and implemented to improve the attractiveness of Kenya as a BPO destination and to encourage investments in this field. This therefore calls for the development of related incentive framework. The framework will define the kind of incentives to be offered to the BPO sector, the stages of the application of the incentives, and the period in which it is to be offered. One such incentive that is already in place is the subsidization of bandwidth costs through financing from the World Bank.

1.3 Computers and computer operating systems

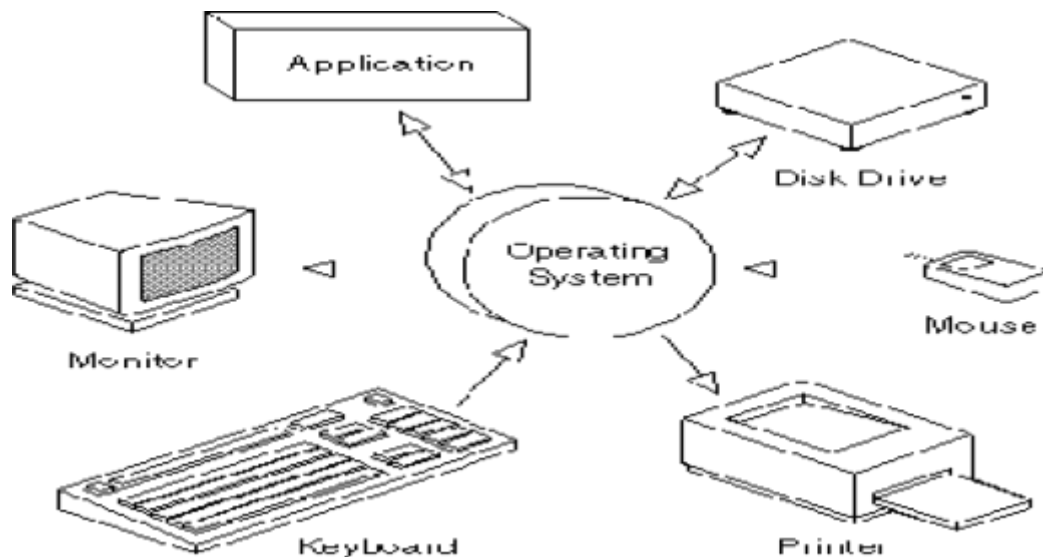
a) What is an Operating System?

- ✦ An operating system is a software program that enables the computer hardware to communicate and operate with the computer software. Without a computer operating system, a computer would be useless.
- ✦ An operating system (commonly abbreviated OS and O/S) is the software component of a computer system that is responsible for the management and coordination of activities and the sharing of the resources of the computer.
- ✦ The operating system acts as a host for application programs that are run on the machine. As a host, one of the purposes of an operating system is to handle the details of the operation of the hardware. This relieves application programs from having to manage these details and makes it easier to write

applications. Almost all computers, including hand-held computers, desktop computers, supercomputers, and even modern video game consoles, use an operating system of some type. Some of the oldest models may however use an embedded OS, which may be contained on a compact disk or other storage device.

- ✦ An operating systems perform basic tasks, such as recognizing input from the keyboard, sending output to the display screen, keeping track of files and directories on the disk, and controlling peripheral devices such as disk drives and printers.

Figure 1.1 Operating Systems



Types of Operating Systems

As computers have progressed and developed, so have the types of operating systems. Below is a basic list of the different types of operating systems and a few examples of operating systems that fall into each of the categories. Many computer operating systems will fall into more than one of the below categories.

(a) GUI - Graphical User Interface

A GUI Operating System contains graphics and icons and is commonly navigated by using a computer mouse. Below are some examples of GUI Operating Systems.

- ✦ System 7.x
- ✦ Windows 98
- ✦ Windows CE

(b) Multi-user

A multi-user operating system allows for multiple users to use the same computer at the same time and/or different times. See our multi-user dictionary definition for a complete definition for a complete definition. Below are some examples of multiuser operating systems.

- ✦ Linux
- ✦ Unix
- ✦ Windows 2000

(c) Multiprocessing

An operating system capable of supporting and utilizing more than one computer processor. Below are some examples of multiprocessing operating systems.

- ✦ Linux
- ✦ Unix
- ✦ Windows 2000

(d) Multitasking

An operating system that is capable of allowing multiple software processes to run at the same time. Below are some examples of multitasking operating systems.

- ✦ Unix
- ✦ Windows 2000

(e) Multithreading

Operating systems that allow different parts of software program to run concurrently.

1.4 Intellectual Property

Intellectual property rights are temporary monopolies enforced by the state regarding use of expressions and ideas.

Intellectual property rights are usually limited to non-rival goods, that is, goods which can be used or enjoyed by many people simultaneously—the use by one person does not exclude use by another. (This is compared to rival goods, such as clothing, which may only be used by one person at a time. For example, any number of people may make use of a mathematical formula simultaneously.)

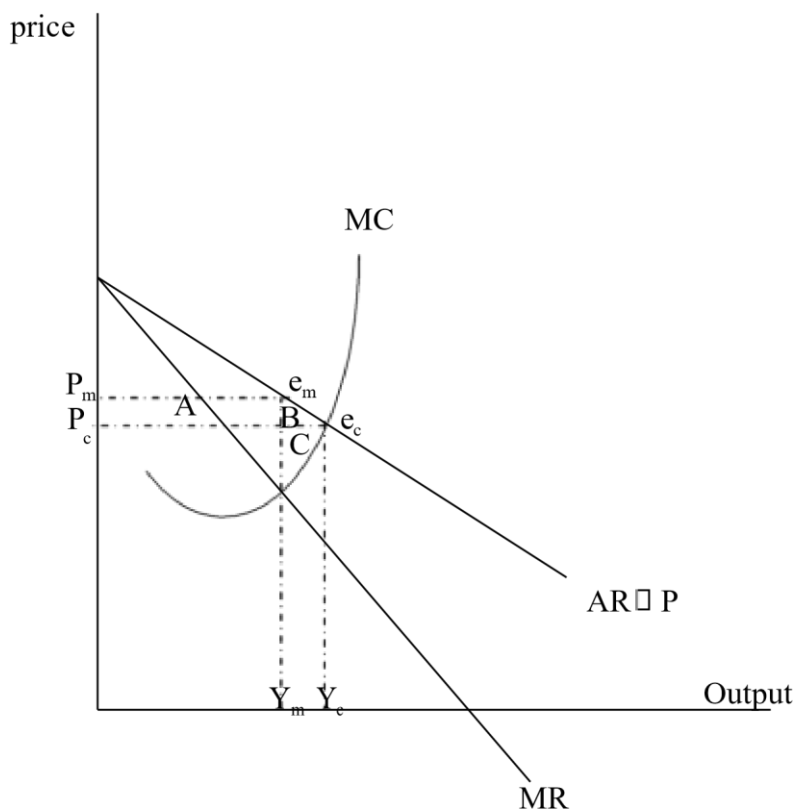
Some objections to the term intellectual property are based on the argument that property can only properly be applied to rival goods (or that one cannot "own" property of this sort).

Since a non-rival good may be used (copied, for example) by many simultaneously (produced with minimal marginal cost), producers would need incentives other than money to create such works. Monopolies, by contrast, also have inefficiencies (producers will charge more and produce less than would be socially desirable).

The establishment of intellectual property rights, therefore, represents a trade-off, to balance the interest of society in the creation of non-rival goods (by encouraging their production) with the problems of monopoly power.

Since the trade-off and the relevant benefits and costs to society will depend on many factors that may be specific to each product and society, the optimum period of time during which the temporary monopoly rights should exist is unclear.

Figure 1.2 Monopoly



2. DATA PROCESSING AND INFORMATION TECHNOLOGY

2.1 Introduction

This widespread and increasingly familiar use of computers, although very important, is not central to Data processing (DP). DP continues to be mainly concerned with primary business operations which require the processing of data to provide information. For convenience we will call this "Conventional DP". In larger organizations many such conventional DP operations continue to be carried out on mainframe computer and minicomputers. It is mostly in smaller organizations that desktop computers have been established as the main vehicle for conventional DP. Nevertheless, this pattern is changing as smaller computers become ever more powerful and are therefore able to support more demanding processing loads thus threatening the status of the larger machines. However, these changes do not

fundamentally alter the continued need to perform conventional DP operations within all organizations.

2.2 Conventional data processing

2.2.1 Basic DP Principles and Practices

An "Information System" is a complete apparatus for handling all aspects of information within an organization. It includes everything from the completely human-oriented aspects of information to the technologically oriented aspects. As we have seen, desktop computing can be effective at providing local computer facilities to IT users particularly with regard to document creation and analysis tasks involving spreadsheets. However, the major centralized components of an information system are the data processing systems within it. It is these that provide much of the information that is needed throughout an organization. Each data processing (DP) system may be designed to fulfill one particular function. Increasingly, however, the DP systems are produced so as to operate as a complete, integrated set of inter-related sub-systems. The more advanced DP systems of this kind are sometimes called "Information Systems", although they clearly do not deal with all aspects of information and therefore the term is not totally merited.

Data Processing (DP) is the term for the process of producing meaningful information by collecting all items of data together and performing operations on them to extract the required information from them. The methods of data processing vary from those that are almost entirely manual to those that rely on the use of large electronic computers. In practice most data processing nowadays will involve the use of some electronic aids such as calculators or small computers, which increase in sophistication year by year.

Examples of Information and Data

The following examples (common to almost all firms) are explained in relation to the information required and the data needed to obtain that information. The

examples have been deliberately selected because they are those with which can (hopefully) readily be identified and used as a yardstick.

- (a) Payroll. Large and small companies have to pay their employees and the information needs are, at least, the payslip (to show the employee how much he or she has earned and the reasons for deductions, etc), a company record of the payslip details (payroll) and analysis for accounting purposes (total tax deducted, etc). If the employees are paid in cash (as opposed to cheque or direct credit to their bank accounts), then a coin analysis may be needed for the bank. For management control, purposes, analysis of labour hours into various categories such as idle time, sickness and absence can be produced. The "data" from which information is derived is likely to have come from some form of time records, e.g. time sheets, and recorded details of employees, e.g. tax codes.
- (b) Invoice. Companies selling goods on credit produce an invoice setting out the details of the sale. These include the customer's name and address, the customer's order number, the date of the sale, and the quantity, price, description and value of the goods sold. This invoice (information) is needed to record the customer's indebtedness. The customer order details and seller's details of good provide the "data" from which it is compiled.
- (c) Statement of Account. Following on from the invoice example, a summary of all the invoices and payments made by the customer in a month, in the form of a "statement", is produced and sent to the customer to show the amount still owing. The statement of Account is the "information" coming from invoices and cash receipts (data).

Factors determining the methods of data processing

Common factors determining the methods of data processing can be explained under the following:

(a) Size and type of organization

(b) Timing aspects

(c) Link between applications

a. Size and type of business

With each of the examples given, the method of producing the information will largely depend on the size and type of business. In a very small company a single person may be able to have the time to produce all the information required, but as the volume of business increases, more people and aids, in the form of calculators and small computers, may be employed. Large volumes of data and information will require the use of large computers.

In some companies the payroll will be a matter of simply paying a member of staff the same amount each month, whilst in others a complex payment by results system will have to be coped with. Similarly, invoicing may be simply a matter of virtually copying from the customer's order, or it may require complex discount calculations. Simple situations indicate the need for fewer people and aid to produce the information and complex situations indicate the need for more people and aids.

b. Timing aspects

Some information requirements are less time critical than others. For example, the Payroll and Statements may only be produced once a month, whereas the invoices may be produced (in certain companies) virtually all the time, i.e. as a customer collects the goods. The timing requirements for information will have considerable bearing on the methods and equipment needed to provide it.

c. Link between applications

Where data is needed for more than one information requirement, a different method of processing it may be suggested. For example, an item sold may not only need to be used in the production of the invoice but also be needed to amend the recorded stock position. A manual system would require separate operations to

satisfy the requirements, whereas a computer system would include the automatic use of data in both applications. This ability of computer systems to perform a variety of processing operations on a single “pool” of data contrasts sharply with manual systems. In manual systems the data being used by one individual becomes inaccessible to another individual.

Data Processing stages

Whatever method or combination of methods is used, it will be seen that data will pass through the same basic stages in the processing cycle. An example is given of the production of a payroll that is simplified for the purpose of illustration but which none the less brings out the salient points.

Payroll example

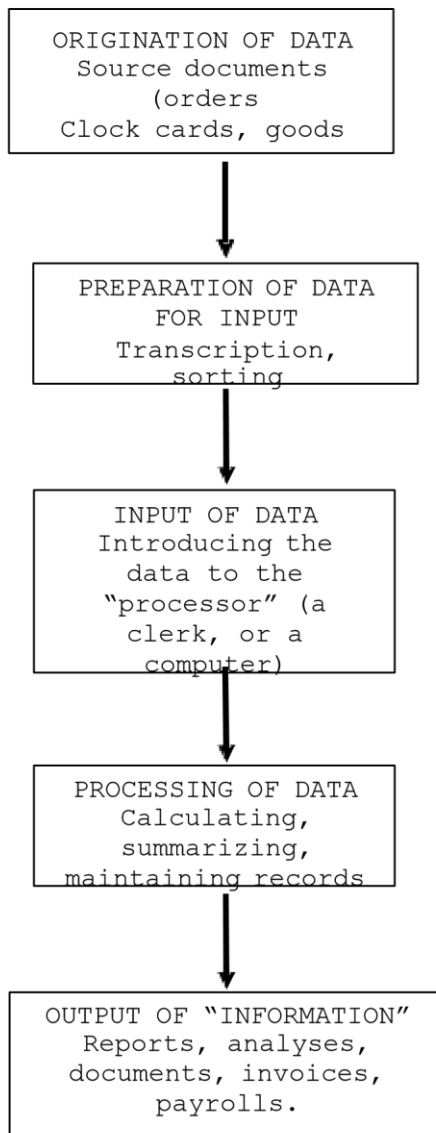
Stage

- i. Details of hours worked by each employee (data (is recorded on a time sheet 9source document). All the time sheets are forwarded to the wages office.
The source documents will be sorted into the sequence required by the payroll.
- ii. The data is then checked for corrections and validity and some form of copying (transcription) onto another document (e.g. summary sheet) may take place. The source documents are then temporarily stored.
- iii. Details of gross pay, insurance and net pay are arrived at. Reference is made to data held in tax tables, in tables of rates of pay and in employee records. The latter must be brought up to date, e.g. pay to Date details.
- iv. Individual pay slips, a summarized payroll and coin analysis are produced (information).

Each of these stages is identified by a name:

- a. Origination – of data
- b. Preparation – getting the data ready

- c. Input – the act of passing the data to the processing stage
- d. Processing – all that is necessary to arrive at the net pay, etc, and to keep data up to date (e.g. data in employee records).
- e. Output – the production of the end product (see figure next page)



Notes:

Data is transmitted from one stage to another. Use will be made of storage at the various stages. Controls will be established where necessary.

The figure above depicts the data processing stages

Note:

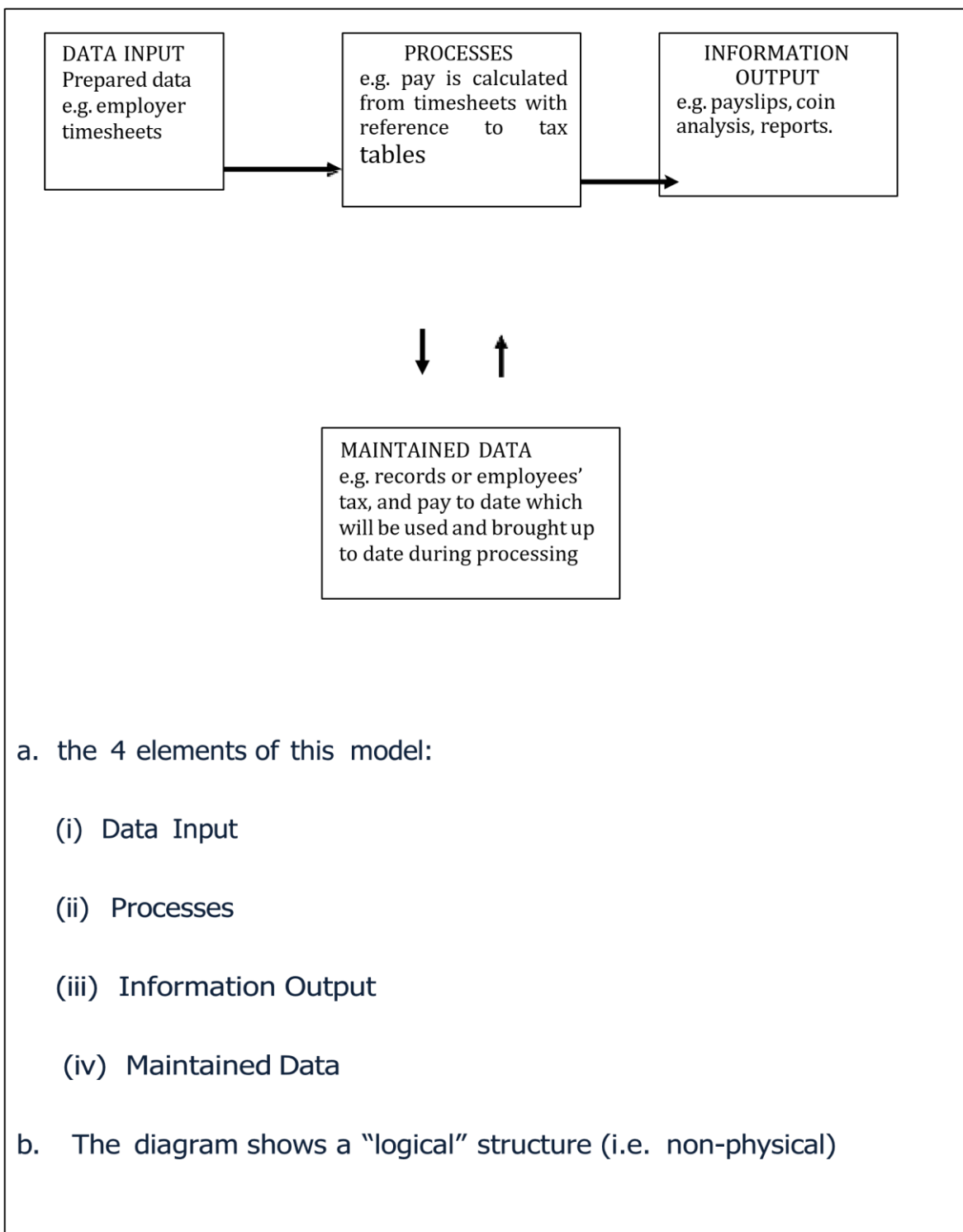
- (a) How data is transmitted from stage to stage
- (b) The use made of temporary storage after preparation

- (c) The reference to stored data (tax tables, rates of pay and employee records) during the processing stage.
- (d) Controls will be part of the whole procedure, e.g. checking time sheets.

DP Model

Having seen a general view of the data processing stages (above figure), we now turn to a general view of the data processing system. In essence all DP systems consist of the four basic interrelated elements illustrated in the figure below. This representation may be referred to as a “model” of the data processing system. It may also be called a “logical model” to emphasize the fact that it is not representing physical features.

This concept of a “model” goes rather deeper and is more useful than this simple illustration suggests. For example, the data within the DP system may be viewed as part of the DP model, in that it represents or reflects the state of affairs, e.g. the current balance represents the state of a customer’s account.



The units of data for processing, such as individual customer order or individual time sheets, are called transactions. Incidentally, the act of processing such a unit of data is also called a transaction, but the context should make it clear which meaning is intended. According to the determining factors, transactions may either be processed singly or in batches. A batch is merely a number of transactions (e.g. in the form of source documents) accumulated together and processed as a single unit.

Batching introduces a time-lag into the processing cycle but also introduces some useful controls e.g. checking that a batch contains the required number of documents helps to detect and correct document loss.

Electronic Data Processing (EDP)

Some form of electronic computer-based data processing system is to be found in almost all except the smallest firm. Thus it is necessary to concentrate our effort into gaining an understanding of what lies behind such systems. Where manual methods of DP are still in use, they frequently form part of an Electronic Data processing (EDP) system and involve the use of simple electronic aids such as calculators. The basic principles of DP and EDP are essentially the same. The computer is merely substituted for manual aids or human labour.

The same DP activities take place, but instead a computer is used to produce the results e.g. the invoices or payroll, etc. Notice that the computer will not take over all of the steps in a given procedure, e.g. the data may first be collected or recorded by conventional methods.

In practice, the computerization of a data processing system entails more than just the automation of parts of the existing system by means of computers. Analysis of an organization's information requirements may show that the requirements will be better served by a newly designed and implemented system with the virtues of both the manual and computerized elements.

What is involved in an EDP system? In order to answer this questions imagine a simple invoicing application.

- (a) The customers' records are held in a magnetic storage device such as a "magnetic disk" or "magnetic tape" (details later). There may not be any visible documentary record. Similarly the records of our items of stock are held in magnetic form.
- (b) When orders are received it is the computer system which is going to process them, but the computer cannot accept them on the pieces of paper the clerk was accustomed to, so we have to convert the source documents into a machine sensible form, e.g. by keying in data at a special keyboard device, or by using machine-readable documents (details later).
- (c) We will have written down all the steps involved in the invoicing procedure and translate. So we now have the sequence of steps in computer language to enable the computer to produce our invoices for us.
- (d) We have:
 - (i) Computer and associated machinery
 - (ii) Master records (customer) in magnetic form
 - (iii) Master records (Stock) in magnetic form
 - (iv) Reference records in magnetic form
 - (v) Transactions (Orders) in by keyboard device or by machine- readable forms
 - (vi) Procedure in computer language

Until the late 1970s the vast majority of EDP activities within most organizations were almost invariably concentrated in specialist DP departments. This was largely a consequence of the cost of computer systems and their specialist staff, and the need to achieve economics of scale. However, conventional DP activities on a small scale, which did not require expensive computer equipment, were often distributed within the various departments of the organizations.

In the early 1980s it became increasingly common to find departments within organizations which had their own small computer-based business computer systems operating quite autonomously of the data processing department. These small computers were often used for a variety of tasks not all of which might be classified as data processing. In addition, the use of other forms of computerized equipment has become widespread, particularly in the areas of office automation

20 and telecommunications. Desktop computing grew in importance. In response to these changes many organizations replaced their data processing department with what are sometimes called "Information Centres". An

Information centre normally contains within it all the traditional functions of a DP department but, in addition, it also has responsibility for a wide range of other information systems services, including the provision and support of small computer systems within the various departments of the organization.

The new technology has been able to establish itself because technological innovations have reduced costs to a point where it can compete with manual methods. The key feature of the new technology is its ability to deal with information in one form or another. The trend has been towards using ways of interconnecting the various types of equipment so as to integrate their functions and manage them more effectively. Thus we see modern EDP taking place not only in specialist DP departments but also being integrated more into the activities of other departments.

In the 1990s technological improvements have reached the point at which the more powerful PCs can challenge the traditional role of minicomputers in terms of performance for a given price and, similarly, minicomputers can challenge the role of mainframes. This, and pressures on organizations to reduce their computing costs, has given rise to an activity known as "downsizing". Downsizing occurs if computer facilities are transferred from mainframes to minicomputers or from minicomputers to microcomputers. As part of this process the strong central controls are removed and the computer facilities become more closely tied to particular business activities. There are clearly two sides to this process. On the positive side are the savings in costs and the greater availability to business users. On the negative side is the loss of useful coherent controls and the potential to needlessly replicate effort and resources. A variant on downsizing is rightsizing which merely means using a combination of computers of all types and sizes with each doing what it does best.

3. ECONOMICS OF SOFTWARE MAINTENANCE

DEFINITIONS

Software

This is a general term used for the various kinds of programs and procedures used to operate computers and related devices.

"Software" is sometimes used in a broader context to mean anything which is not hardware but which is used with hardware

The hardware describes the physical aspects of a computer and related devices
e.g. Monitor, mouse, keyboard, CPU

TYPES OF SOFTWARE

System software

This is a software used to control the computer and helps to develop and run applications. It includes programs such as operating systems, database managers, compilers, Web servers, router operating systems and control programs for other network devices. E.g. windows XP

Firmware

This is a software program or set of instructions programmed on a hardware device. They are fixed small programs and data structures that internally control various electronic devices. Examples of devices containing firmware include remote controls, calculators, mobile phones, digital cameras etc

Firmware is involved with very basic low-level operations in a device, without which the device would be completely non-functional.

Application Software

This is a computer software designed to help the user perform a particular task. Examples include word processors, spreadsheets, media players, database applications e.t.c.

Programming software

This is a software that usually provides tools to assist a programmer in writing computer programs, and softwares using different programming languages in a more convenient way.

PROGRAMMING LANGUAGES

A programming language is an artificial language designed to express computations that can be performed by a machine, particularly a computer. Programming languages can be used to create programs that control the behavior of a machine.

They are the basic building blocks for all software, allowing people to tell computers what to do and the means by which systems are developed.

There are over 3000 programming languages. Examples include C, C++, Visual basic, JAVA e.t.c

SOFTWARE MAINTENANCE

Software maintenance has many interpretations. The two most common meanings of the word software maintenance include:

1. Defect repairs;
2. Enhancements - adding new features to existing software applications.

There are 21 major kinds of work performed under computer maintenance

- a) Major Enhancements – adding new features with over 20 function points

- b) Minor Enhancements – adding new features with over 5 function points
- c) Maintenance - repairing defects for good will
- d) Warranty repairs - repairing defects under formal contract
- e) Customer support - responding to client phone calls or problem reports
- f) Error-prone module removal - eliminating very troublesome code segments
- g) Mandatory changes - required or statutory changes
- h) Complexity analysis - quantifying control flow using complexity metrics)
- i) Code restructuring - reducing any essential complexity
- j) Optimization - increasing performance
- k) Migration - moving software from one platform to another
- l) Conversion - Changing the interface or file structure
- m) Reverse engineering - extracting latent design information from code
- n) Reengineering - transforming legacy application to client-server form
- o) Dead code removal - removing segments no longer utilized
- p) Dormant application elimination - archiving unused software
- q) Nationalization - modifying software for international use
- r) Year 2000 Repairs - date format expansion or masking
- s) Euro-currency conversion - adding the new unified currency to financial applications
- t) Retirement - withdrawing an application from active service
- u) Field service - sending maintenance members to client locations

All the 21 maintenance topics are different in many respects. However, they all involve modifying an existing application rather than starting from scratch with a new application.

MEASUREMENT OF MAINTENANCE AND ENHANCEMENT ACTIVITIES

Each of the 21 works produces different levels of results when maintaining software. The difference or variability depends on

- 1) Technical factors,

2) Experience levels of the maintenance personnel.

How are the results measured?

Metrics is used to measure the results of software maintenance.

Definition

In software development and maintenance, a metric is the measurement of a particular characteristic of a program's performance or efficiency.

Two useful metrics can be used to estimate the various maintenance tasks;

1) Assignment scope

It refers to the amount of software one programmer can keep operational in the normal course of a year, assuming he/she is doing routine defect repairs and minor updates

2) Production rate

It refers to the number of units that can be handled in a standard time period such as a work month, work week, day, or hour. Production rates are mostly expressed in terms of function points (units) per staff per month.

Definition

A function point is a unit of measurement to express the amount of business functionality an information system provides to a user.

Economics in software maintenance

In economics, the aim is optimization. Profits are maximized by minimizing cost.

← □ TR TC

One of the major costs to the producers is labour cost. An organization with a computerized system that supports their daily operations will have to incur costs to maintain it.

If a company owns a portfolio of 100,000 function points maintained by generalists many more people will be required than if maintenance specialists are used. This will increase cost.

Example: A company has used complexity analysis tools and has a staff of highly trained maintenance specialists. The company has maintenance assignment scope of 3,000 function points. This company might only need 30 maintenance experts as opposed to 200 generalists.

Metrics Problems with Small Maintenance Projects

There are several difficulties in exploring software maintenance costs with accuracy.

- 1) Maintenance tasks are often assigned to development personnel who combine both development and maintenance as the need arises. This practice makes it difficult to distinguish maintenance costs from development costs because the programmers are often rather careless in recording how time is spent.
- 2) A great deal of software maintenance consists of making very small changes to software applications. Some repairs may involve fixing only a single line of code. Adding minor new features such as adding a new line- item on a screen may require very few statements (source code statements). These small changes are below the effective lower limit for counting function point metrics. It is difficult to count function points below a level of 15 function points.
- 3) With a very small project, one staff can repair a function point per work day. If the project takes one work day consisting of six hours, then at least the results can be expressed using common metrics. In this case, the results would be roughly "6 staff hours per function point. This brings inefficiency in resource allocation.

Best Practices in Software Maintenance

Because maintenance of aging legacy software is very labor intensive it is quite important to explore the best and most cost effective methods available for dealing with the millions of applications that currently exist.

The practices that have the most positive impact on maintenance are;

- a) Utilization of full-time, trained maintenance specialists rather than turning over maintenance tasks to untrained generalists.
- b) Use of staff with high experience
- c) Use of low complexity of base code – use of simple source codes or understandable computer programming language
- d) Use of proper tools for each task – for each of the 21 works
- e) Ensuring quality and constant appraisal and measurement
- f) Formal base code and software inspections
- g) Excellent response time by maintenance experts
- h) Annual training for users and repairers
- i) HELP desk automation
- j) No error prone modules - A portion of a program that carries out a specific function and may be used alone or combined with other modules of the same program.
- k) On-line defect reporting
- l) Productivity measurements
- m) Excellent ease of use
- n) User satisfaction measurements
- o) High team morale

Worst Practices in Software Maintenance

Factors which exert a negative impact on the work of updating or modifying existing software applications include;

- a) Having Error prone modules
- b) Staff inexperience
- c) High complexity of base code
- d) Low level programming languages
- e) No defect tracking tools
- f) Lack of Y2K "mass update" specialists
- g) Poor ease of use
- h) No quality measurements and controls
- i) Lack of maintenance specialists
- j) Poor response time
- k) Management inexperience
- l) No base code inspections
- m) No HELP desk automation
- n) No on-line defect reporting
- o) No annual training
- p) No maintenance tools
- q) No productivity measurements
- r) Poor team morale
- s) No user satisfaction measurements

Software Entropy and Total Cost of Ownership

Entropy means the tendency of systems to destabilize and become more chaotic over time. All known compound objects decay and become more complex with the passage of time unless effort is exerted to keep them repaired and updated.

In software, the accumulation of small updates over time tends to gradually degrade the initial structure of applications and makes changes grow more difficult over time.

If applications are developed with minimal initial quality control they will probably be poorly structured and contain error-prone modules. This means that every year, the accumulation of defect repairs and maintenance updates will degrade the

original structure and make each change slightly more difficult. Over time, the application will destabilize and ad fixes will increase in number and severity. Unless the application is restructured or fully refurbished, eventually it will become so complex. Maintenance will now only be performed by a few experts who are more aware of the changes in the application.

On the other hand, leading applications that are well structured initially can delay the onset of entropy. Well-structured applications can achieve declining maintenance costs over time. This is because updates do not degrade the original structure, as happens low graded softwares.

The total cost of ownership of a software application is the sum of four major expense elements:

- 1) The initial cost of building an application
- 2) The cost of enhancing the application with new features over its lifetime
- 3) The cost of repairing defects and bugs over the application's lifetime
- 4) The cost of customer support for fielding and responding to queries and customer-reported defects.

4. ECONOMIC BENEFITS OF INFORMATION TECHNOLOGY REVOLUTION

In the new global economy information and communications technology (ICT) is the major driver of improved quality of life for people and of economic growth.

THE DIGITAL ECONOMY

- The digital economy represents the wide use of IT (hardware, software, applications and telecommunications) in all aspects of the economy, including internal operations of organizations (i.e. in businesses, government and non-profit organizations). - IT can be used in;
 - ✦ Transactions between individuals
 - ✦ Transactions between organizations,
 - ✦ Transactions between individuals and organizations
- The technology underlying the digital economy goes far beyond the Internet and personal computers. IT is embedded in a vast array of products including cell phones, MP3 players, digital cameras, washing machines, cars, and credit cards, and industrial machines and tools, lasers, and robots.
- As IT continues getting cheaper, faster, better, and easier to use, organizations continuously find new and expanded uses for it every day.
- IT is a general purpose technology. A general purpose technology has three characteristics;
 - ✦ It is pervasive – it is used widely by most sectors in the economy.
 - ✦ The performance and price improve with time
 - ✦ It makes the invention and production of new products, processes and business models easy

Due to these characteristics, IT has transformed many economies through improved productivity and efficiency. This has led to economic growth in these economies.

INFORMATION TECHNOLOGY AND PRODUCTIVITY

Productivity is a measure of output from a production process, per unit of input.

In economics, there are two main measures of productivity.

Labor productivity - The amount of output a worker achieves given a particular unit of work.

Capital productivity - This is the measure of how well physical capital is used in providing goods and services.

Example,

If a company driver using a cell phone can deliver more packages to clients a day than a driver without, the labor productivity of the former driver is higher than the latter. However in this case, the former worker is using more technology, which the company must pay for.

How do we control for changes in inputs, such as more physical capital (e.g., cell phone) or more human capital (e.g., more training for the driver)?

Total factor productivity is used. Higher total factor productivity is an indication of greater efficiency in the economy because of better technology or economic organization as opposed to simply more capital.

IT boosts both labor productivity and total factor productivity. Higher productivity is important in ensuring robust economic growth.

Cobb-douglas production function

$$Q = AK^aL^b$$

Productivity growth - This is the increase in the amount of output produced by workers per a given unit of effort.

At both the firm and the country/national level, greater investment in IT is associated with greater productivity growth.

Productivity in Firms and in the Economy

Studies have found that the more firms invest in IT the higher their productivity. Productivity is significantly higher in plants running sophisticated software to integrate multiple business processes, such as inventory and production.

E.g. In the USA it has been found that the use of computer networks raises productivity in manufacturing plants by about 7.5 percent.

In the UK it has been found that an additional 10 percent of workers using computers result in a 2.2 percent gain in productivity in older firms and 4.4 percent in new firms.

There are two kinds of productivity effects from technology;

- ✦ Capital deepening - as workers get more capital they are generally more productive.
- ✦ Increases in total factor productivity (TFP) - occur when the same amount of capital is used to produce more goods/services

There are three reasons why IT boosts productivity and has stronger effects on productivity than other capital;

- ✦ IT equipment innovations are new and are relatively easy to improve efficiencies. This is particularly true in the service sector where before the IT revolution it was difficult to use existing capital equipment (largely electro-mechanical machines) to boost productivity.
- ✦ IT automates tasks and also has widespread complementary effects. This allows companies to reengineer and restructure their processes.
- ✦ IT has spillover effects. These are realized by adding additional users to a network. Increasing the user size of a network makes all current users better off. For example, the first e-mail applications were not practical because so few people had e-mail. But as more organizations provided their workers with e-mail and connected to the Internet, e-mail became valuable and essential.

IT and Workers Productivity

How does IT boost worker productivity?

- ✦ IT lets workers do more things at the same time
- ✦ IT lets workers focus more exclusively on valuable work and avoid the less productive distractions that are part of many jobs.
- ✦ IT allows routine tasks to be automated, thereby increasing economic output. – e.g. standing orders
- ✦ IT enables organizations to dramatically improve the efficiency of internal operations.
- ✦ IT helps government streamline routine and often costly transactions.

- ✦ IT lets bits to be substituted for paper. Processing paper, plastic and other physical forms of information is quite expensive compared to processing digital bits
- ✦ IT enables more productive self-service

INFORMATION TECHNOLOGY AND ECONOMIC GROWTH

IT contributes to economic growth through the following channels;

- It enables creation of larger markets
- It gives managers better tools by which to make decisions, thereby improving firm performance (if firms are doing well, the economy grows)
- It enables more people to work thus boosting economic output

Larger Markets

IT revolution has enabled the emergence of more globalized markets in both manufacturing and services.

Examples;

- Use of Visa and other banking services enables transactions across borders.
- Computer-based logistics systems allow companies to weave together multinational supply-chains
- e-mail and cheap telecommunications systems allow managers to easily communicate across the globe

IT allows companies to achieve greater economies of scale and exerts greater competitive pressure on firms to boost performance.

Better decision making

IT enables organizations to make processes more efficient. Better faster accurate decisions are made.

IT is creating real-time enterprises. This is where enterprises are able to react instantaneously to changes in business.

IT has enabled the creation digital dashboards. This is a software for executives that bring minute-by-minute updates of market conditions and firm vital signs right to the executive's computer.

IT also enables better decision-making by allowing decision makers to use data to find patterns and predict outcomes.

Increased economic output

IT helps boost economic output by making it easier for people to work who otherwise could not. This includes people who have more work flexibility and people with disabilities who previously would have limited their involvement in the labor market.

IT enables self employment and people can work from home. In this way individuals with responsibilities can care for dependents (either children or aged parents) and work part-time from home.

There are computer products for people with disability. E.g. a newly developed Braille printer by ViewPlus Technologies prints in Braille and text so that both visually impaired and sighted people can read the same manuscript and collaborate better.

Assignment

What is IT bust?

INFORMATION TECHNOLOGY AND BUSINESS CYCLES

A business cycle is a pattern of alternating periods of economic growth (recovery) and decline (recession), characterized by changing employment, industrial productivity, and interest rates.

IT dampens the severity of the business cycle. This is through two ways;

- ✦ Financial innovation
- ✦ Improved inventory management

Financial Innovation

Technological advances have made it easier for lenders to collect and disseminate information on the creditworthiness of prospective borrowers. Lending institutions are able to better assess and price risk. Many firms and individuals who would not have had access to credit can qualify. This means productive businesses are funded and banks collect back their money without default. In this case, individuals and businesses are able to smooth their consumption through tough time thus the economy does not suffer major downturns.

Improved inventory management

IT has reduced instability in inventories. The build-up of inventory, particularly in durable goods industries like autos, steel, and appliances, has been a major cause of economic slowdowns. Because of the inflexibility of production systems, coupled with the difficulty in assuring fast replacement of inventories, companies have a propensity to build up high levels of inventory. When this happens, companies cut production in order to sell off some inventory. They make fewer products and buy

less from suppliers, leading workers to being laid off, which dampens consumer demand, and which in turn cut sales even more. This in turn leads to more layoffs and the downward cycle becomes severe.

IT makes the production system operate more in real time. Products are made without stocking. The dynamism of the products does not also allow inventories. Future levels of demand are predicted accurately thus reducing inventories and business cycles.

INFORMATION TECHNOLOGY AND EFFICIENCY

Efficiency is producing the right quantity of goods and selling them at the right price as dictated by the forces of demand and supply. In this case, you cannot make someone better off without making someone else worse off.

The digital economy helps to make the allocation of goods and services more efficient.

Well-functioning markets are the principal way to ensure high levels of allocation efficiency. But well-functioning markets depend on at least three important factors:

- ✦ Prices that reflect costs
- ✦ Well-functioning exchange markets,
- ✦ Good information by buyers and sellers.

In many cases, markets lack some or all of these factors. IT remedies these deficiencies by;

- ✦ It boosts allocation efficiency by enabling the creation of markets and market signals where before there were none.

e.g. parking in town by hours

- ✦ It helps markets be more efficient by expanding consumer information. Buyers need information to make the best possible choices. Knowing what products are available, how they are priced, and the precise nature of those products is critical for finding the best combination of goods and services for one's money. The internet offers all this.

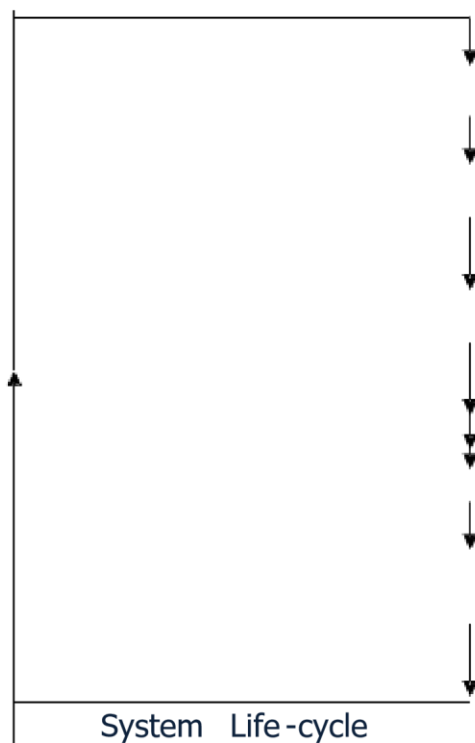
5. INFORMATION SYSTEMS DEVELOPMENT

Development Frameworks

The new computer systems frequently replace existing manual systems, and the new systems may themselves be replaced after some time. The process of replacing the old system by the new is often organized into a series of stages. The whole process is called the system life cycle. The system life cycle may be regarded as a "framework" upon which the work can be organized and methods applied to develop a new system.

The Systems Life Cycle:

The life cycle can be broken down into separate stages. The figure below shows the major stages in a typical system life cycle.



This development approach is referred to as waterfall method because of the Preliminary survey or study

Feasibility study Investigation

and fact recording Analysis

Design

Implementation

way in which what is produced at each stage cascades down to the stage below.

The start of a new system life cycle is normally due to factors such as; Perception of a business need.

- Failure/limitations of the existing system causing dissatisfaction
- Heightened awareness of modern developments

Due to the factor(s) above, the management will initiate the selection of a project for preliminary study or investigation.

- i) Preliminary survey or study: To establish whether there is a need for a new system, and of so to specify the objectives of the system.
- ii) Feasibility study: To investigate the project deeply to be able to provide information that either justifies the development of the new system or shows why the project should not continue. Report is presented to the management with appropriate recommendations.
- iii) Investigation and fact recording: A detailed study is conducted to fully understand the existing system and to identify the basic information requirements. This requires contributions of the users of both the existing system and the proposed system.
- iv) Analysis: Analysis of the full description of the existing system and of the objectives of the proposed system should lead to a full specification of the user's

requirements. This requirements specification are examined and approved before system design is embarked upon.

- v) Design: This depends on the analysis, e.g. different combinations of manual and computerized elements may be considered.
- vi) Implementation: Involves following the details set out in the system specification. Three particular tasks here are hardware provision, programming and staff training. The implementation stage is sometimes split into a 'build phase' and 'test phase'.
- vii) Maintenance and Review: Implemented system is examined to see if it has met the objectives set out in the original specification. Unforeseen problems may need to be overcome by returning to the earlier stages in the cycle to take corrective measures.

Development Methods

There are traditional and modern methods used in systems developments. However, there are some basic principles that apply to whatever method of system development. These basic principles are expressed as a set of recommendations, as follows;

- 1) Make sure you understand the problem.
- 2) Fully identify the requirements and get them agreed by the client.
- 3) Clearly document all work as you do it.
- 4) Have all work checked to an agreed standard of quality at every stage.
- 5) Be systematic.
- 6) Be creative but make sure it works.
- 7) Allow for future changes.
- 8) Make sure that all users find the system usable.
- 9) Break the problem down into manageable tasks and carry them out in a planned and methodical way.
- 10) Make sure that the final system is all there, in full working order, and does what it's supposed to do, before you say it's ready.

(Note: Students should be able to explain these points)

Modern methods:

Below are two examples of modern methods of system development.

- (i) Data Models: is a representation of properties of the data within an existing or proposed system. The process of constructing a data model from scratch is called data analysis. Entities, Attributes and Relationships (EAR) Models are the most common and successful types of data model. The model is made up of;

- Entity – Anything about which data can be stored, e.g. customers or products.
- Attribute – Facts that need to be stored about the entity, e.g. customer's account number, name, address, etc.
- Relationship – The relationships that exist between various entities in the system, e.g. there may be a relationship between customer and an order. For such a model to be considered valid it must conform to a set of rules. A valid data model is one which is fully normalized; the process of validating a data model is called normalization. The aim of normalization is to ensure that each fact is only recorded in one place so that facts cannot be inconsistent and performance of updates cannot produce anomalies by updating one copy of the fact but not another.

Each occurrence of an entity must be uniquely identifiable by means of a key containing one or more attributes, e.g. customer's full name or account number might serve as a key. So given the key, other facts relating to the key can be obtained.

- (ii) Data-Flow Diagrams (DFDs): Provides an effective form in which to present the movement of data through system and associated transformations of the data resulting from various processing actions taken upon it. DFDs provide a view which is free from unnecessary details and are therefore very useful in providing an overview of the system.

Traditional Methods:

A number of traditional methods are still important and are often incorporated into the more successful modern methodologies;

- Term of reference: Clear objectives must be laid down by management for all projects. Analysts must not set their own objectives. Analysts must therefore

conform to the terms of reference before beginning their task. These may include the standards which must be used in the project.

Fact finding: It is essential to gather all facts about a current system to ensure that all strengths and weaknesses are discovered, and the new system designed to eliminate the weaknesses while retaining the strengths. Techniques available for fact finding include; Interviewing, Questionnaires, Observation, Record Inspection and User Workshops.

Fact recording: The investigator must formulate a plan of keeping notes of the facts obtained during the fact finding stage. At this stage, no attempt is made at analysis or design since the existing system must be fully understood first.

Form design: Design of forms and design of procedures are very much linked. The completion of a form may be the first operation of a procedure, (e.g. a receipt, as a by product of posting to the cash book, in a machine accounting). The aim of form design is to keep forms to the minimum consistent with serving the needs of the system. The form design may have the following features;

(a) Sizes: standard size forms should be used, its more economic to do so, and handling, filing and copying are simplified.

(b) Types of paper: Should be appropriate considering the frequency of handling, storage needs, conditions under which form are completed and prestige requirements.

(c) Identification: There should be brief, self explanatory title, copies identified by different colours or bold symbols, serial numbering for internal check purposes.

(d) Common information: If two forms are used in conjunction with each other, the common information should be in the same sequence and position.

(e) Vertical spacing: There should be adequate spacing for each item of entity.

(f) Columns: The length of column heading should be tailored to the width required by the information to be entered in the column.

(g) Pre-printing: Common details should be pre-printed leaving only variable data to be entered.

(h) Clarity: Instructions should be simple. Material of similar nature grouped together. Logical sequence for completion should be followed.

- (i) Miscellaneous: There may be a requirement for perforation for subsequent bursting or pre-punched holes for subsequent filing.
- (j) Multi-part sets: Where more than one document is to be raised at the same time, consideration should be given to the method employed in carrying the image through all copies, e.g. no-carbon-required paper, carbon patches, etc.

Internal check: The arrangement of procedures and systems, and the allocation of duties so that there is an automatic check on what is being done. No one person should be placed in the position of having responsibility for all aspects of a business transaction. The work flow is arranged so that the work of one person/section is independently checked by some other person/section. Internal checks helps in the following ways;

- (i) Reducing possibility of fraud and error
- (ii) Determining the responsibility for errors
- (iii) Supervision
- (iv) Reducing the work for both internal & external auditors. Examples of internal checks include; serial numbering of forms, two signature requirement on cheques and use of control accounts.

Analysis: After the analyst has gathered all facts which are relative to the objectives set and has grouped them into a form suitable for analysis, he/she must examine all facts gathered in order to make proper assessment of the existing system. The aim of this stage to ensure that all feasible alternatives are considered. The present system may be criticized against on the following principles of procedure

- (i) Purpose – are the purposes being satisfied? Are they still necessary? Could they be achieved in other way?
- (ii) Economical – is it economical? Benefits & costs.
- (iii) Work flow – are the work flows satisfactory? (Subject of renewed interest in the recent years).
- (iv) Specification/simplification/standardization - are the three S'' being practical? Can the work be computerized? Can complex procedures be simplified?

- (v) Flexibility – is the system flexible? Can it handle big increase or decrease in volumes to be processed?
- (vi) Exception principle – is the principle of exception being observed?
- (vii) Reliability
- (viii) Form – is the information being produced in the form best suited to the recipient? Is there a need for a hard copy/
- (ix) Existing system – if change is made, what equipment and other facilities currently being used could be incorporated in the new procedure?
- (x) Continuous control – what types of errors are occurring? Are controls satisfactory/ what other types of controls could be used?
- (xi) Time – is the information being produced in the time for meaningful action to be taken?

Implementation and Post-implementation

Implementation follows on from the detailed design stage. It involves the coordination of the efforts of the user department and the data-processing department in getting the new system into operation. A coordinating committee is sometimes formed for this purpose, composed of managers from the concerned departments, representatives from the computer department and the analysts responsible for the design of the new system.

Planning for implementations begins at the design stage, and includes the following;

- Training of staff – The amount of training required for various categories of personnel will depend upon the complexity of the system and the skills presently available. Appropriate aids like handbooks, courses and lectures would be required.
- Programming – The programmer must design programs that conform to the requirements set out in the system specification.
- System testing – To ensure that both individual programs have been written correctly and that the system as a whole will work, i.e. link between the programs in a suite.
- Master file conversion – it is necessary to convert master files into magnetic form. The stages of file conversion will depend on the method currently used for keeping the files, e.g. manually, in box files.

Changeover procedures – there are two basic methods of changing over to a new system;

- (i) Parallel – The old and new systems are run concurrently, using same inputs. The outputs are compared and reasons for differences solved. The outputs from the old system continue to be distributed until the new system has proved satisfactory; the old system is now discontinued.
- (ii) Direct – The old system is discontinued and the new system becomes operational immediately. There must be complete confidence in the new system's reliability and accuracy before the method is used.

However, there is a case referred to as 'pilot' where by a variation of the two main methods is used. This is possible when part of the system can be treated as a separate entity, e.g. department or branch might be computerized before the undertaking as a whole.

Review and Maintenance – the examination of the new system to see if it has met its objectives, e.g. costs and benefits are compared with the estimates produced at the system's inception. This is often referred to as 'post-audit'. The system is reviewed and maintained periodically to;

- (i) Deal with unforeseen problems arising in operations.
- (ii) Confirm that planned objectives are being met and take necessary action if not.
- (iii) Ensure that the system is able to cope with the changing requirements of business

Operations after implementation – the day to day operations of the computer system rely upon a number of specialist staff such as operations manager and computer operators. Other types of specialists include; computer/DP manager, data-control staff, data preparation staff, tape/dist librarian, computer supervisor.

Project Management

Good planning is a prerequisite of any business activity and computer systems development is no exception. Planning and management of systems development projects traces the steps involved in the system life cycle, hence system

development project management primarily concerns the management aspects of the system life cycle. Assuming the new system involves computerization in a company which is not currently using computers for the applications concerned, the steps would be;

- Preliminary survey/study – the board must carry out investigations into the possibility of using the new computer system. This is done by senior executives or consultants. Application areas are considered in order of priority. The outcome of the preliminary study will determine whether or not to continue investigating the need for a computer. If so, specific computer objectives are drawn up.
- Steering Committee – when a project is established, a steering committee for the project is also established to keep the project on course so that it meets its objectives. The membership of the committee comprises of senior management members and senior project staff such as analysts, so that the committee can carry out monitoring, controlling and consultative functions.
- Feasibility study – if the project is given go-ahead, a study team is formed to carry out a study of the application area(s). This team works under the steering committee, and provides information on the technical, economical and social aspects of the project. The steps in the study depends on individual circumstances but generally include the following;
 - (a) The team be aware of the precise objectives of the study.
 - (b) Establish the requirements of the particular system being studied. (c) Examine the system to determine hoe the objectives can be met, e.g. information flow, time scale, etc.
 - (d) Note any improvements that can be made in the system as it stands as these may well be implemented at once.
 - (e) Formulate the new design on the basis of application of computer methods. Manual methods can also be used where appropriate. (f) Assess costs of possible new system (staff, building, hardware, software, etc) over the life of the project.
 - (g) Make comparisons with present system.
 - (h) Considerations of social factors, i.e. effect on the staff relations, etc.

(i) Make recommendations to the board in the form of a Feasibility Study Report.

Feasibility study must be carried out in enough depth to enable the board to make a decision on whether or not to proceed with detailed investigation and analysis. The content of the report should provide information on aim and objectives, cost/benefit comparison, outline of proposed system, effects on the organization, schedule of requirements, implementation and recommendations.

Consideration of report – the board will consider the report and will want answers to the following questions, Do the recommendations meet the objective?

Do they fit overall company objectives?

What will be the effect on company profits?

What will be the effect on company organization and staff?

What are costs involved and is the capital available?

What is the time scale for implementation?

Does it look sufficiently into the future, i.e. does the proposal allow for possible future expansion and modification?

The board may ask for further information in certain areas. After the report has been studied in detail, a decision has to be taken whether or not to proceed.

Detailed investigation and analysis – if the decision taken is to proceed, this stage continues the process started with the initial survey and the feasibility study, and a requirement specification is produced. This requirement specification provides the basis for a go/no-go decision between the feasibility study report and the full system specification.

Consideration of the requirement specification – a part from the answering the questions above (under consideration of report), the board will expect a satisfactory answer on whether the specification give a full and accurate statement of the data processing requirements. Although the requirement specification is 'implementation free' it contains sufficient detail for the main design alternative to be identified. The board can approve the specification and decide whether or not to proceed with one or more designs.

Facility selection – selection of one of the main design alternatives is effectively a selection of the kind of facilities to be used, e.g. manufacturers

and suppliers of a particular product may be selected and asked to submit proposals that indicate how they can meet the requirements. Alternatively, an independent consultancy firm may be requested to handle the problem.

- Design – the outcome of this stage is the ultimate in documentation of the new system, the system specification. This will contain full details of all clerical and computer procedures involved. Each system specification is considered, along with any associated proposals from the manufacturers or suppliers, and one of the alternatives is selected and pursued.
- Resource planning – detailed planning and preparation need to be done in order that the installation of the machines and changeover to the computer-based system are achieved smoothly and efficiently; Network of related activities and events must be put in place, main resources (site, building, environment, standby equipment, etc) arranged, staff are selected and trained (managerial, clerical, analysts and programmers, operators) and finances needed are made available.
- Installation of computer – manufacturer's Engineers will be responsible for installation of the machinery in conjunction with the computer manager or his representatives. The machine is tested and handed over. Maintenance of hardware and software will normally be the subject of an agreement and will ensure the machine is always in working order and spare parts available when required.
- Implementation – this consists of; training of staff, programming, system testing, master life conversion, changeover procedures, review (system evaluation) and maintenance.
- Program writing and testing – programmers get on with the job of producing the computer programs from the systems specification. Individual procedures are programmed, compiled and tested for logical correctness using 'dummy' data. After being tested individually, the programs are tested as a complete system. Programmers must ensure that the program meets its specification. The programs should be subjected to independent testing, by people other than the program writers.
- Review (system evaluation) and maintenance – once implementation process is complete and the system is operational, it is important to carry out evaluation process to ensure the original objectives are being met. This

is sometimes called 'post-audit'. The findings of this operation will always help in future projects. For the new system to run efficiently, it must be constantly monitored and maintained. Maintenance is bound to be necessary sooner or later and should be planned for.

6. DIFFERENTIATION OF PRODUCTS AND PRICES

Product differentiation

Product differentiation is the process of distinguishing a product from others, to make it more attractive to a particular target market. Products are considered to be differentiated when there are physical differences or attributes which may be real or perceived by buyers so that one product is preferred over that of a rival firm. Products are differentiated by firms in order to obtain higher prices or increased sales or both.

Price discrimination

This is whereby the seller sells different units of output at different prices. Price discrimination is important in high tech industries for two reasons;

1. The high-fixed cost, low-marginal-cost technologies commonly observed in these industries often leads to significant market power, with the usual inefficiencies. In particular, price will often exceed marginal cost, meaning that the profit benefits to price discrimination will be significant to the participants.
2. Information technology allows for a keen observation and analysis of consumer behaviour. This allows for various kinds of marketing strategies that were previously extremely difficult to carry out, at least on a large scale. A seller can offer prices and goods that are differentiated by individual behaviour and/or characteristics.

There are three kinds of price discrimination.

1. First-degree price discrimination

This is the most extreme case. Information technology allows for a "market of one," in the sense that highly personalized products can be sold at a highly

personalized price. This phenomenon is also known as "mass customization" or "personalization."

Consumers can personalize products. They can buy a personally configured computer from hp, Dell, or even purchase customized blue jeans from Levi's.

Presumably companies will find it much more attractive to fine-tune pricing in Internet-based commerce, eliminating ``menu costs" from the pricing decision. Internet retailers revise their prices much more often than conventional retailers, and their prices are adjusted in much finer increments.

The theory of monopoly first-degree price discrimination argues that firms will charge the highest price they can to each consumer, thereby capturing the entire consumer surplus.

Limitation

This is an extreme case. On-line sellers face competition from each other and from off-line sellers. Adding competition to this textbook model is important.

Two effects

First-degree price discrimination has two significant effects:

1. The enhanced surplus extraction effect - refers to the fact that personalized pricing allows firms to charge prices closer to the reservation price for each consumer (the reservation (or reserve) price is the maximum price a buyer is willing to pay for a good or service).
2. The intensified competition effect - refers to the fact that each consumer is a market to be contested. When consumer tastes are not dramatically different, the intensified competition effect dominates the surplus extraction effect, making firms worse off and consumers better off with competitive personalized pricing than with non-personalized pricing.

Advantages

Personalized pricing obviously raises privacy issues. A seller that knows its customers' tastes can sell them products that fit their needs better but it will also be able to charge more for the superior service.

2. Second-degree price discrimination

Second-degree price discrimination refers to a situation where everyone faces the same menu of prices for a set of related products. It is also known as "product line pricing," "market segmentation," or "versioning." The idea is that sellers use their knowledge of the distribution of consumer tastes to design a product line that appeals to different market segments.

This form of price discrimination is widely used. Automobiles, consumer electronics, and many other products are commonly sold in product lines.

Information technology is helpful in both collecting information about consumers, to help design product lines, and in actually producing the different versions of the product itself.

Limitation

The basic problem in designing a product line is "competing against yourself." Often consumers with high willingness to pay will be attracted by lower-priced products that are targeted towards consumers with lower willingness to pay. This "self-selection problem" can be solved by lowering the price of the high-end products, by lowering the "quality" of the low-end products, or by some combination of the two.

Advantages

Versioning is good in that it allows markets to be served that would otherwise not be served. This is the standard output-enhancing effect of price discrimination.

3. Third-degree price discrimination

Third degree price discrimination is selling at different prices to different groups. It is a classic form of price discrimination and is widely used.

The conventional treatment examines monopoly price discrimination, but there have been some recent attempts to extend this analysis to the competitive case. It has been observed that when consumers have essentially the same tastes, and there is a fixed cost of servicing each consumer, then competitive third-degree price discrimination will generally make consumers better off.

Competition forces firms to maximize consumer utility, and price discrimination gives them additional flexibility in dealing with the fixed cost. If there are no fixed costs, consumer utility falls with competitive third-degree price discrimination.

Conditioning on purchase history

Another form of price discrimination that is of considerable interest in high-tech markets is price discrimination based on purchase history. A monopolist can discriminate between old and new customers by offering upgrades and enhancements. In a duopoly model which adds an additional phenomenon of ``poaching:" one firm can offer a low price to steal another's customers.

SEARCHING PRICE INFORMATION

IT through the Internet can lower the cost of search quite dramatically. Even in markets where there are relatively few direct transactions over the Internet, consumers gather information before purchase. There are many shopping agents that allow for easy price comparisons.

However sellers use a mixed strategy and randomize the prices they charge. This allows them to sometimes charge low prices so as to compete for the searchers and still charge, on average, a high price to the non-searchers.

Disadvantage of searching

One reason that many people don't use search engines is that they do not trust the results. It is common for online retailers to engage in bad tactics i.e. they advertise an inferior version of a product in order to attract users to their site.

BUNDLING

Bundling refers to the practice of selling two or more distinct goods together for a single price. This is particularly attractive for information goods since the marginal cost of adding an extra good to a bundle is negligible. E.g. Microsoft office has packages (word, excel, access, power point, publisher, outlook)

There are two distinct economic effects involved;

- ✦ Reduced dispersion of willingness to pay, which is a form of price discrimination
- ✦ Increased barriers to entry

7. SWITCHING COSTS AND LOCK-IN

When you switch automobiles from Toyota to Mazda, the change is relatively painless. However, switching softwares is very expensive. Switching from Windows to Linux can be very costly.

You will have to change document formats, applications software, and you will have to invest substantial time and effort in learning the new operating environment.

Changing software environments at the organizational level is also very costly. Changing the operating system involves additional costs of infrastructure upgrades, consultants, and retraining programs.

These switching costs are endemic in high-technology industries and can be so large that switching suppliers is virtually impossible. This is a situation known as “lock-in.”

Analytics of Lock-in

Consider the following simple two-period model.

There are n consumers, each of whom is willing to pay v per period to buy a nondurable good. There are two producers that produce the good at a constant identical marginal cost of c . The producers are unable to commit to future prices. In order to switch consumption from one firm to the other, a consumer must pay a switching cost s . We suppose $v \geq c$, but $v + s < c$, so that it pays each consumer to purchase the good but not to switch. The unique Nash equilibrium in the second period is for each firm to set its price to the monopoly price v , making profit of $v - c$. The seller can extract full monopoly profit second period, since the consumer are “locked-in,” meaning that their switching costs are so high that the competitive seller is unable to offer them a price sufficiently low to induce them to switch.

Competition to acquire customers

When switching costs are substantial, competition can be intense to attract new customers, since, once they are locked in, they can be a substantial source of profit.

Examples:

- (1) One can buy a nice ink jet printer for \$150 only to discover a few months later that the replacement cartridges cost \$50. The notable fact is not that the cartridges are expensive, but rather that the printer is so cheap. And, of course, the printer is so cheap because the cartridges are so expensive. The printer manufacturers are following the time-tested strategy of giving away the razor to sell the blades. That is, the printer manufacturers believe that since the customer has bought the printer (of course cheaply), he/she will have to buy the cartridges anyway – the consumer is locked in. Business Week reports that in 2000, HP's printer supply division made an estimated \$500 million in operating profit on sales of \$2.4 billion.

The rest of HP's businesses lost \$100 million on revenues of \$9.2 billion. The inkjet cartridges reportedly have over 50% profit margins

- (2) Switching costs in the credit card and bank loan markets are also substantial: in the bank loan case, switching costs amount to about a third of the average interest rate on loans.

As these examples illustrate, lock-in can be very profitable for firms. It is not obvious that switching costs necessarily reduce consumer welfare, since the competition to acquire the customers can be quite beneficial to consumers. For example, consumers who use their printers much less than average are clearly made better off by having a low price for printers, even though they have to pay a high price for cartridges.

On the other hand, some firms (e.g. Microsoft or Oracle) suffer from the "burden of the locked-in customers," in the sense that they would like to sell at a high price to their current customers (on account of their switching costs) but would also like to compete aggressively for new customers, since they will remain customers for a long time and contribute to future profit flows. This naturally

leads such firms to want to price discriminate in favor of new customers, and such strategies are commonly used.

Though in many cases welfare may go either way, it is found that switching costs are generally bad for consumer welfare: they typically raise prices over the lifetime of the product, create deadweight loss, and reduce entry.

Analytics of competition to acquire customers

Continuing with the model considered under 'simple analytics of lock-in' above, suppose for simplicity that the discount rate is zero, so that the sellers only care about the sum of the profit over the two periods. In this case, each firm would be willing to pay up to $v - c$ to acquire a customer. Bertrand competition pushes the present value of the profit of each firm to zero, yielding a first period price of $2c - v$. The higher the second-period monopoly payoff, the smaller the first period price will be, reflecting the result of the competition to acquire the monopoly. If we assumed the goods were partial substitutes, rather than perfect substitutes, we would get a less extreme result, but it is still typically the case that the first-period price is lower because of the second-period lock-in. It is worth noting that the conclusion that first-period prices are lower due to switching costs depends heavily on the assumption that the sellers cannot commit to second-period prices. If the sellers can commit to second period prices, the model collapses to a one-period model, where the usual Bertrand result holds. In the specific model discussed here, the price for two periods of consumption would be competed down to $2c$.

Switching costs and price discrimination

One common example of switching costs involves specialized supplies, as with inkjet printer cartridges. In this example, the switching cost is the purchase of a new printer. The market is competitive ex ante, but since cartridges are incompatible, it is monopolized ex post. This situation can also be viewed as a form of price discrimination. The consumer cares about the price of the printer plus the price of however many cartridges the consumer buys. If all consumers are identical, a monopolist that commit to future prices would set the price of the cartridges equal to their marginal cost and use its monopoly power on the printer.